Three consecutive sums of two squares

Putnam, 2000 A2

Theorem 1. *There are infinitely many integers n such that n, $n+1$, and $n+2$ are each the sum of two squares of integers.*

Proof. To prove that there are infinitely many such integers, it suffices to show that for every integer N there exists an integer $n > N$ with the required property.

Let N be an arbitrary integer, and set

$$t = |N| + 1.$$

Then $t \geq 1$, and since $N \leq |N|$, we have

$$N < t.$$

Moreover,

$$(2t)^2 - t = 4t^2 - t = t(4t - 1) \geq 0,$$

because $t \geq 0$ and $4t - 1 \geq 0$. Hence

$$t \leq (2t)^2.$$

Since $(2t^2)^2 \geq 0$, we also have

$$(2t)^2 \leq (2t^2)^2 + (2t)^2.$$

Consequently, if we set

$$n = (2t^2)^2 + (2t)^2,$$

then

$$N < t \leq (2t)^2 \leq (2t^2)^2 + (2t)^2 = n,$$

so $n > N$.

It remains to write n, $n + 1$, and $n + 2$ as sums of two squares. First, by definition,

$$n = (2t^2)^2 + (2t)^2.$$

Next,

$$n + 1 = (2t^2)^2 + (2t)^2 + 1 = 4t^4 + 4t^2 + 1 = (2t^2 + 1)^2 + 0^2.$$

Finally,

$$n + 2 = 4t^4 + 4t^2 + 2 = (2t^2 + 1)^2 + 1^2.$$

Thus n, $n+1$, and $n+2$ are each the sum of two squares of integers, and $n > N$. Since N was arbitrary, there are infinitely many such integers n. □